Response to the Referee Report

Symmetry control of exceptional structure in the massive scalar quasinormal spectrum of doubly rotating five-dimensional Myers–Perry black holes

[manuscript number]

We thank the referee for their reading of the manuscript. Changes in the revised version are marked in blue. Below we reproduce each point in italics and respond beneath it.

Anticipated points

The following are objections we expect and for which the evidence already exists in the manuscript or the public repository. They are listed so that the response can be specific rather than defensive.

Referee: *The result is negative; how do you know you simply did not look hard enough, or that your detector could not fire?*

Response: The detector is validated on closed-form problems with known exceptional structure ($\lambda^2 - t$, $\lambda^3 - t$, Jordan blocks) *and* on negative controls where it must stay silent (avoided crossings, semisimple and block-diagonal degeneracies). See Supplemental Material, Sec. S4. Separately, the exclusion is quantitative: three rescaling-invariant diagnostics are bounded away from zero, so the statement is a bound rather than an absence of sightings.

Referee: *Exceptional lines are known in Myers–Perry–de Sitter. Is your result in conflict?*

Response: No. That mechanism relies on the cosmological horizon and the near-Nariai Pöschl–Teller limit, neither of which has a $\Lambda = 0$ analogue. This is stated in Sec. X of the manuscript and in the novelty audit in the repository.

Referee: *Why not simply use $|dF/d\omega|$ as the exceptional-point diagnostic?*

Response: Because it is not invariant: the spectral condition is defined up to a nonvanishing analytic factor, so $|dF/d\omega|$ at a root can be rescaled at will. We exhibit two conditions with identical zeros whose $|dF/d\omega|$ differ by more than 10^5. See Sec. V.

Referee: *The minimum gap sits on the boundary of your search box.*

Response: Correct, and stated. A probe beyond the box shows the gap falling further at larger s but rising again at larger μ, so it does not extrapolate to zero; we make no claim outside the declared domain.

Referee: *Can the no-go be promoted to the unseparated problem / to a continuum theorem?*

Response: The no-go already *is* a statement about the unseparated pencil: the decomposition is over (m_1, m_2) only and uses the torus action, not separation of variables in θ. It is not a numerical statement and carries no discretization. Certification of the *numerical* results is a separate matter and is deliberately limited; see Supplemental Material, Sec. S5.

Referee report

Referee: ***[referee point 1]***

Response: **[response]**

Referee: ***[referee point 2]***

Response: **[response]**